User Behavior Analysis and Purchase Prediction in an Online Store

Big Data Final Project | Innopolis University | Team 18

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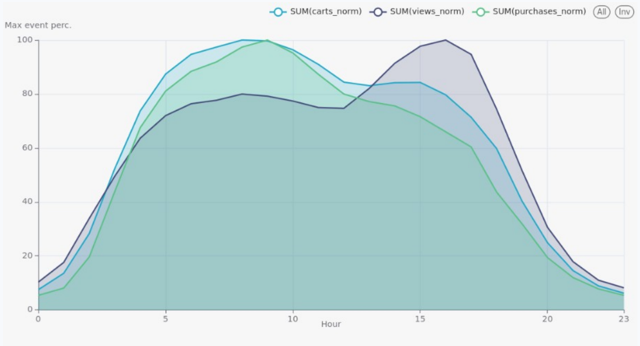

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Presentation roadmap

15-20 minute defence structure

Business problem and project objective
Dataset scale and user-event structure
End-to-end big data pipeline
Six EDA insights from Superset dashboards
Spark ML purchase prediction task
Dashboard, results, recommendations, and challenges

Storyline: from raw events to business decisions

Collect → Store → Analyze → Predict → Present

The goal is not only to process data at scale, but to explain how users browse, abandon carts, and purchase.

Project objective

Analyze ecommerce behavior and predict purchase sessions

Business question

How can an online store use large-scale event data to increase conversion, reduce abandonment, and allocate marketing budget more efficiently?

When are users most likely to buy?
Which categories and brands deserve more support?
Which users are passive browsers vs. high-value customers?
Can we predict purchase from the first 10 session events?

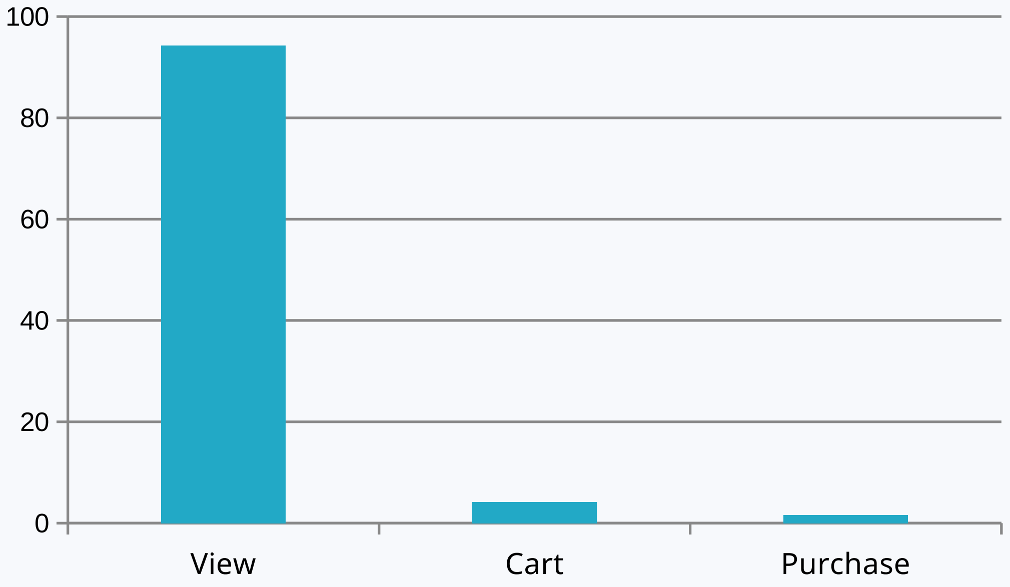
Output

Superset dashboard with data characteristics and six insights
Spark ML models trained on YARN
Model metrics and prediction example
Actionable business recommendations

Dataset at scale

Kaggle ecommerce behavior data, October-December 2019

177.5M	7.86M	38.6M	239.7K	4,846
events	unique users	sessions	products	brands



Key characteristic: views dominate the dataset, while purchases are rare and business-critical.

This imbalance makes PR AUC important for model evaluation because it focuses on the positive purchase class.

End-to-end big data pipeline

From raw CSV files to dashboard and ML outputs

Raw Dataset → PostgreSQL/Citus → Sqoop → HDFS → Hive → SQL on Tez → Spark ML on YARN → Superset

Data collection was automated with Kaggle API download and extraction.
PostgreSQL staging stored the ecommerce events before HDFS ingestion.
Sqoop imported tables as Avro files with Snappy compression.
Hive prepared optimized analytical tables for EDA and ML.
Spark ML trained distributed models on YARN, not local mode.

Main deliverable

A reproducible big data workflow with stage scripts, Hive outputs, ML artifacts, and Superset dashboard.

Data storage and preparation

Optimized Hive warehouse for analytics

Hive table	Format	Compression	Optimization
events_raw	Avro	Snappy	External table over Sqoop output
events_partitioned	Parquet	Snappy	Partitioned by year/month/day
session_features	Parquet	Snappy	Partitioned by date, bucketed by user_id

Why this matters

Parquet + Snappy improves storage efficiency and query speed.
Partitioning supports time-based analytics.
Bucketing by user_id supports session and user-history feature generation.
EDA and ML use the same warehouse layer, making the pipeline consistent.

EDA dashboard overview

Six business insights before predictive modeling

1. Daily shopping rhythm

When users browse and buy

2. Category performance

Top purchase categories and niche opportunities

3. Customer segmentation

High-value customers vs browsers

4. Brand loyalty

Apple/Samsung loyalty and repeat buyers

5. Session funnel

Where sessions drop before purchase

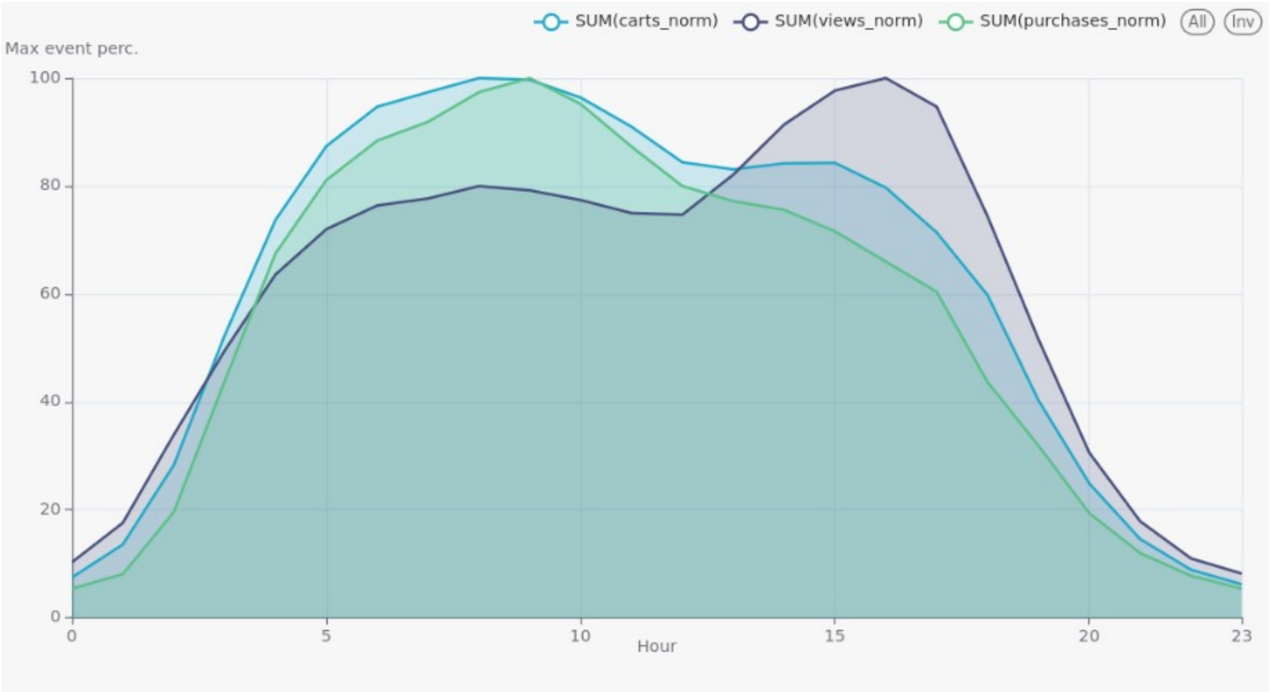
6. Weekday vs weekend

High-intent weekend mornings

Each insight follows the same logic: business question → chart → data fact → interpretation → recommendation.

Insight 1 - Daily shopping rhythm

Traffic peaks and purchase intent do not fully coincide



Finding

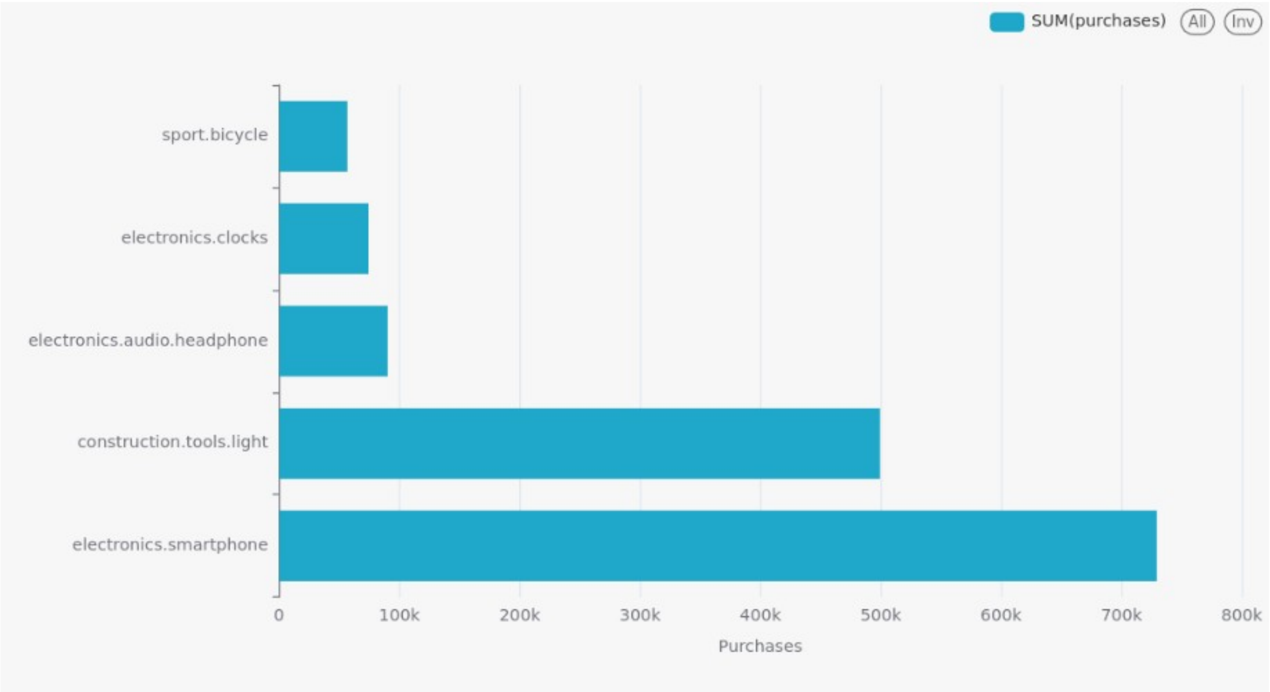
Browsing activity rises during the morning and peaks in the afternoon.
Purchase intent is stronger earlier in the day.
Evening users may generate higher-value purchases.

Action

Schedule conversion campaigns in the morning/early afternoon; use evening for premium offers and bundles.

Insight 2 - Category performance

A few product categories dominate purchases



Finding

electronics.smartphone is the strongest volume driver.
construction.tools.light is another major purchase category.
Smaller categories can still show strong cart-to-purchase efficiency.

Action

Protect inventory and visibility for top categories; test growth campaigns for high-conversion niche categories.

Insight 3 - Customer segmentation

High-value customers behave very differently from browsers

Finding

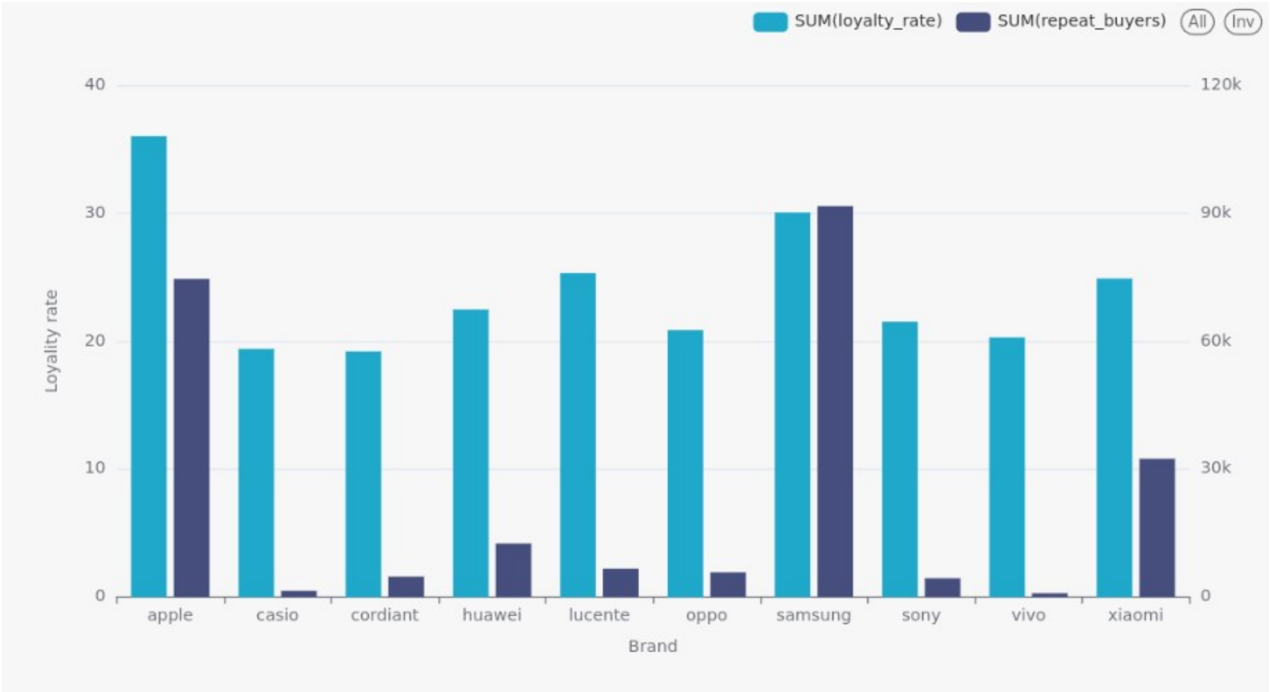
Conversion is concentrated among high-value and regular customers. Browsing-oriented segments generate activity but little purchase behavior. User groups require different lifecycle strategies.

Action

Activate passive browsers, nudge engaged browsers, and protect high-value customers with VIP retention programs.

Insight 4 - Brand loyalty and repeat buyers

Brand-level behavior reveals partnership opportunities



Key numbers

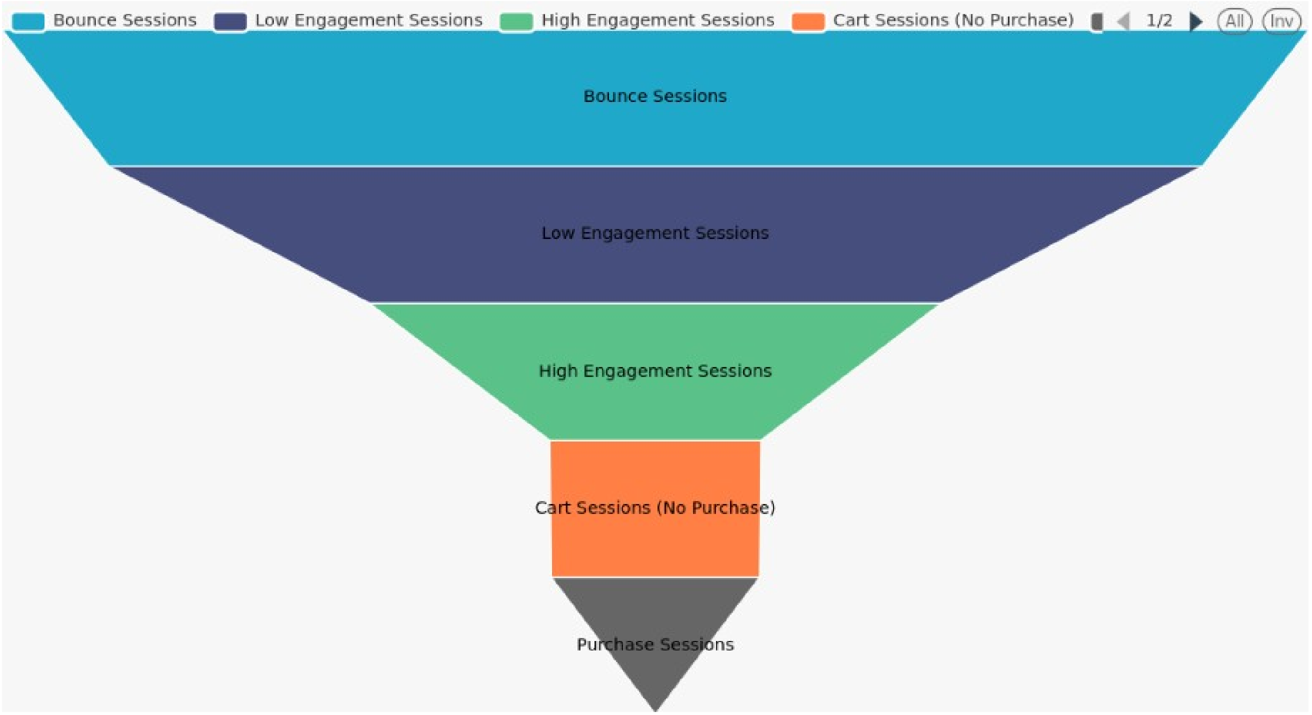
Samsung: 638,864 purchases; 30.05% loyalty rate.
Apple: 518,449 purchases; 36.02% loyalty rate.
Repeat buyers indicate stable demand and higher customer value.

Action

Prioritize partnerships with loyal brands; use upgrade programs, accessories, and bundles to increase lifetime value.

Insight 5 - Session funnel

Cart abandoners are the most valuable near-conversion group



Finding

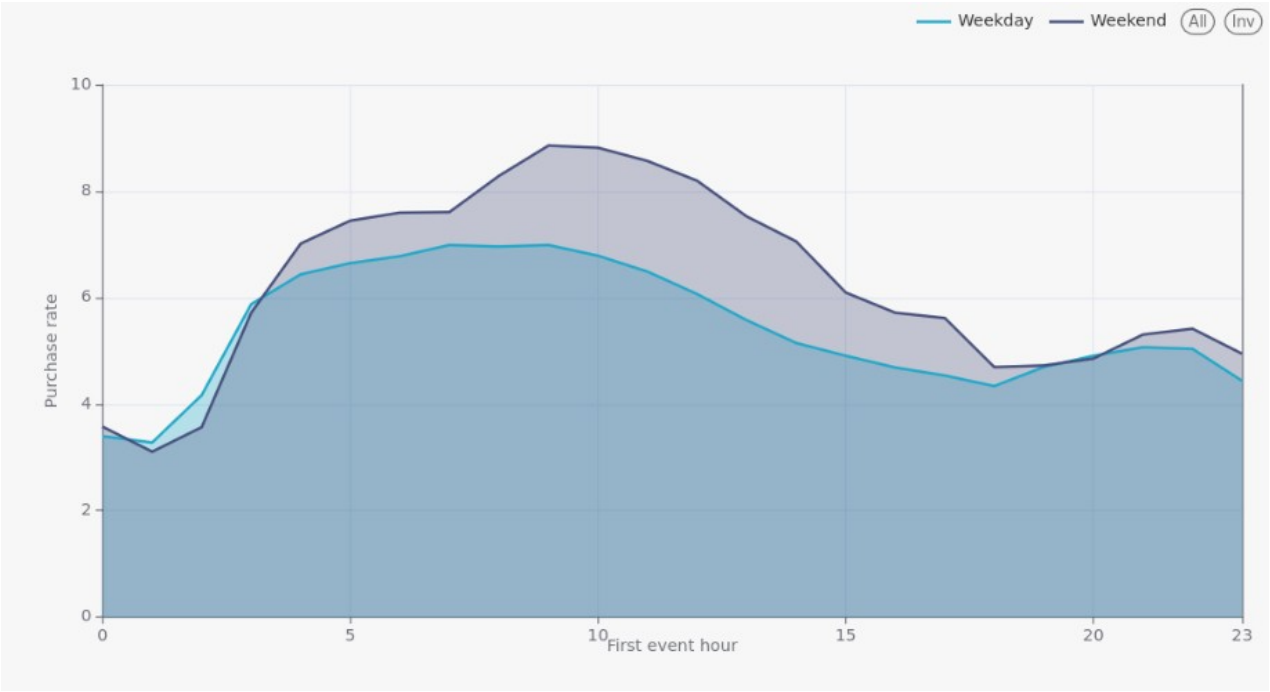
Many sessions end before meaningful engagement.
Purchase sessions are the smallest funnel segment.
Cart sessions without purchase are close to conversion.

Action

Improve landing/product discovery for early drop-off and trigger real-time recovery for users who add to cart.

Insight 6 - Weekday vs weekend

Weekend mornings are a high-intent purchase window



Finding

Weekend purchase rate is higher in the morning.
Weak weekday afternoon periods are less efficient.
Time-based behavior is useful for campaign scheduling.

Action

Concentrate flash sales and paid campaigns during weekend mornings; use weekday afternoons for retargeting.

Predictive analytics task

Predict whether a session will end with purchase

Binary classification

Label = 1: purchase occurs after the first 10 events
Label = 0: no purchase occurs

Prediction uses early-session behavior only.
This prevents leakage and simulates real-time intervention.
Models were trained as distributed Spark DataFrames on YARN.

Why first 10 events?

After a short prefix, the platform can decide whether to show a recommendation, reminder, or cart incentive before the user leaves.

Use case: real-time purchase probability score

Feature engineering and training setup

From raw events to session-level ML features

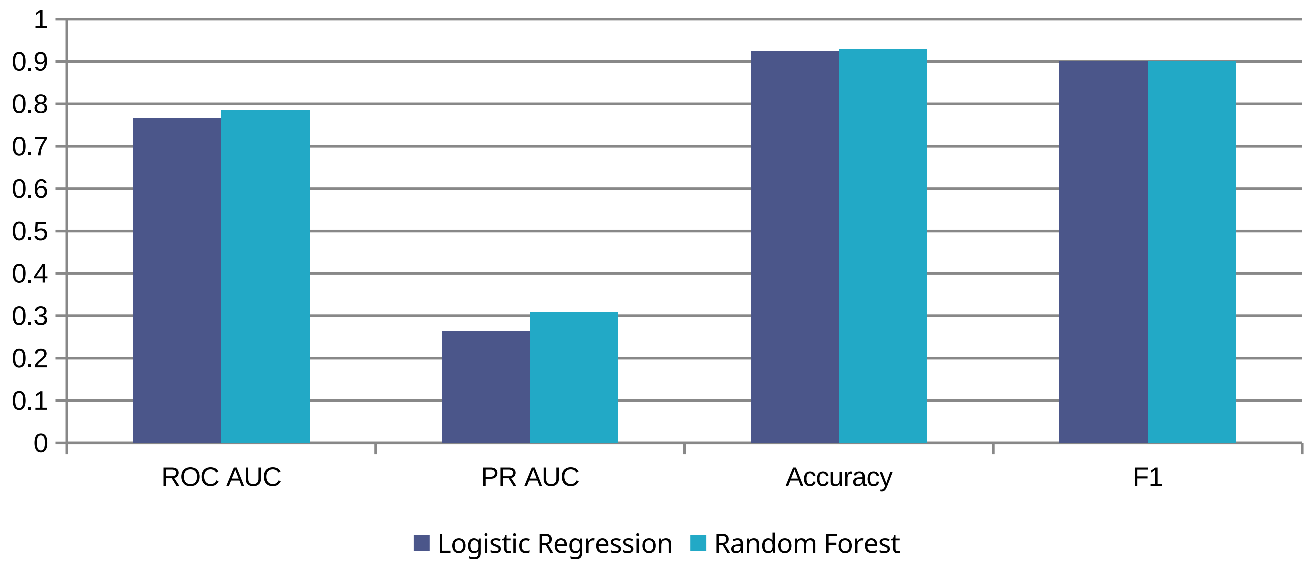
Prefix counts: views, carts, removed carts, distinct products/categories/brands.
Price and pace: average price, max/min price, prefix duration, time between events.
Funnel ratios: cart-to-view rate, remove-to-cart rate, cart in first 3 events.
User history: 30-day purchases, days since last session, user purchase rate.
Temporal encodings: hour/day/month sin and cos transformations.
Categorical prefix context: last prefix event type with StringIndexer + OneHotEncoder.

Training setup: 70% train / 30% test · 3-fold cross-validation · grid search · test set used only for final evaluation

Model comparison

Random Forest selected as the best model

Model	ROC AUC	PR AUC	Accuracy	F1
Logistic Regression	0.7662	0.2627	0.9249	0.9006
Random Forest	0.7845	0.3085	0.9283	0.9001



Interpretation

Random Forest has higher ROC AUC, PR AUC, and Accuracy.
F1 is comparable for both models.
PR AUC is important because purchases are a minority class.
The selected model can support targeted interventions.

Prediction example

A session with purchase after the first 10 observed events

Input prefix behavior

10 prefix events between 04:55:00 and 04:57:44.
Mostly product views within the same category.
The 10th event is a cart action for product 12201687.
After the prefix, the session contains purchases at 04:58:00 and 04:59:13.

Business meaning

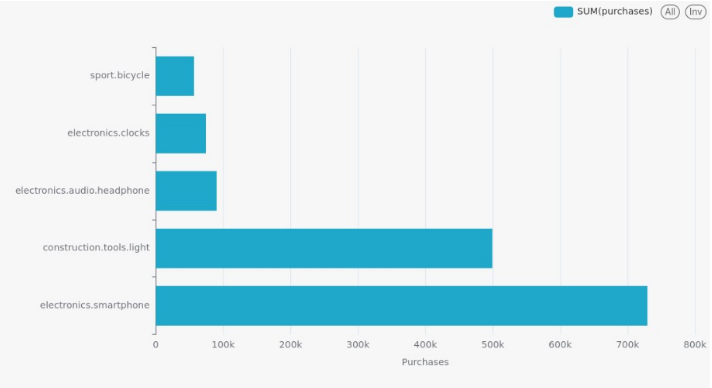
This is exactly the moment where the platform can intervene: the user is already showing purchase-like behavior before the final purchase happens.

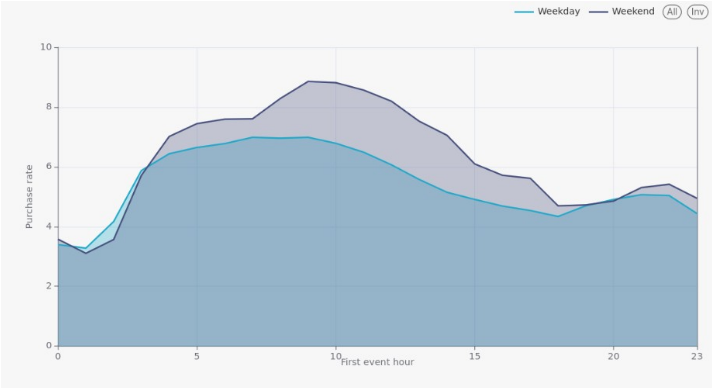
Real-time action examples

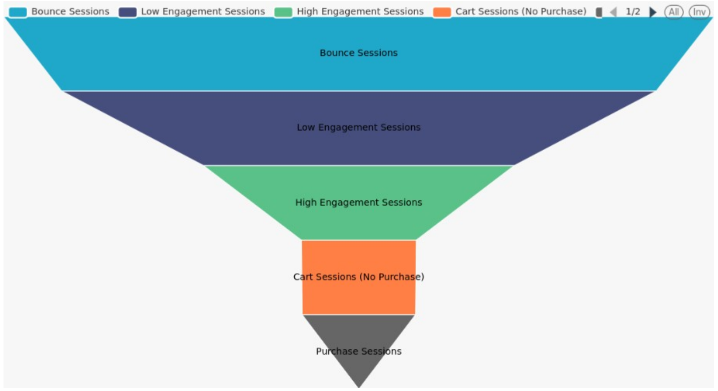
Personalized recommendation
Free delivery reminder
Small cart incentive
Urgency message

Apache Superset dashboard

One place for data characteristics, EDA, ML metrics, and predictions







Dashboard answers

When to schedule campaigns?
Which categories and brands should be prioritized?
Where do users drop off?
Which model predicts purchase sessions best?

Challenges and engineering decisions

What made the project difficult

Dataset source instability: direct Rees46 links were unavailable, so the pipeline used Kaggle API download.
Large-scale ETL: stages were split into reproducible scripts with explicit logging.
Storage/query performance: Hive tables were redesigned to Parquet + Snappy with partitioning and bucketing.
Feature leakage risk: labels are based only on events after the first 10 prefix events.
Integration: EDA and ML outputs had to be dashboard-ready for Superset.

Conclusion and business recommendations

From big data processing to actionable ecommerce decisions

- Schedule conversion campaigns during morning and weekend-morning high-intent windows.
- Use evening periods for premium offers and bundles.
- Protect inventory and advertising visibility for top categories.
- Create activation campaigns for passive browsers and VIP retention for high-value customers.
- Trigger real-time cart recovery for near-purchase sessions.
- Use purchase probability scores to personalize offers and reduce wasted marketing spend.

Final takeaway

The project shows how a Hadoop/Spark/Superset pipeline can transform raw ecommerce events into business insights and predictive tools.